

An Introduction to CRISPR-Mediated Genome Editing in Fungi [†]

Valmik K. Vyas¹ and Douglas A. Bernstein^{2*}

¹Whitehead Institute for Biomedical Research, Cambridge, MA 02142

²Department of Biology, Ball State University, Muncie, IN 47306;

Central dogma, transformation, and genome editing are key biological concepts for which junior scientists must gain experience during training. Here we present an exercise that introduces these concepts in a single practical laboratory exercise. Our exercise utilizes CRISPR/Cas9 genome editing to introduce a stop codon into *Saccharomyces cerevisiae* ADE2. This edit leads to the buildup of an adenine precursor that dyes the edited cells red. As the repair template, guide RNA, and Cas9 are all encoded in our vector, transformation can be performed in 2 hours. Furthermore, since all components of the Cas9/CRISPR system are encoded by the vector, specialized equipment and reagents, such as a PCR machine or oligonucleotides, are not required to perform the experiments as designed. As such, these exercises provide an efficient cost-effective introduction to a wide variety of key molecular biology concepts and lay the foundation for more rigorous investigations in upper-level classes and independent research projects.

INTRODUCTION

The central dogma of biology states DNA is transcribed to RNA, which is then translated to protein (Fig. 1) (1). As such, the sequence of DNA—or genotype—informs the proteins that are to be made, thus influencing phenotype. Genetic modification or mutagenesis of an organism genotype is one tool scientists use to investigate phenotype (1). One way to perform targeted mutagenesis is genome editing using Cas9, which is part of the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) system (2). In bacteria, CRISPR systems form a type of immune system, which defends the host by cleaving invading genetic material, such as viruses (3). Scientists are leveraging the power of CRISPR systems to edit the genomes of a wide variety of organisms. Application of this technology is fundamentally changing both industrial and academic science (4). To edit a genome with CRISPR, an investigator needs three things: 1) a nuclease that will cut the genome, most commonly Cas9; 2) a 20-base guide RNA (gRNA), which will guide Cas9 to the specific genome sequence using base pairing; 3) a repair template that, in coordination with the host DNA repair machinery, directs the repair of the double-strand break

introduced by Cas9 and, in doing so, introduces a mutation (edit) to the genome (Fig. 2) (1).

The precision of this technique comes from the specificity of the 20-base guide RNA, which binds to complementary DNA bases at the target site. Since there are four possible bases in DNA (represented as A, G, C, and T), there are 4²⁰ different 20-base pair sequence combinations. It is easy to change the sequence CRISPR targets by changing the sequence of the guide RNA. For cutting to happen at any given 20-base pair sequence, there is one additional requirement—the presence of a 3' protospacer adjacent motif (PAM) sequence (Fig. 2). For Cas9, the PAM sequence is NGG, where N is any base (A, G, C, or T) (5, 6). After the Cas9/gRNA complex finds its corresponding DNA binding site, cleavage occurs on both DNA strands. Scissors indicate cleavage sites (Fig. 2). The cell's DNA repair machinery uses the repair template to fix the double-stranded break. The repair template must disrupt the Cas9/gRNA nuclease recognition elements (gRNA binding site and/or PAM) for genome editing to occur. If the recognition elements were to remain, the nuclease could potentially bind and cleave the target sequence again.

PROCEDURE

Overview

In this exercise, students edit the genome of the budding yeast *S. cerevisiae* (7). People have been using yeast for thousands of years for making bread, wine, and chocolate—in recent years, it has also proven useful for the production

*Corresponding author. Mailing address: Department of Biology, Ball State University, Muncie, IN 47306. Phone: 765-285-8851. E-mail: dabernstein@bsu.edu.

Received: 3 April 2019, Accepted: 26 August 2019, Published: 18 December 2019.

[†]Supplemental materials available at <http://asmscience.org/jmbe>

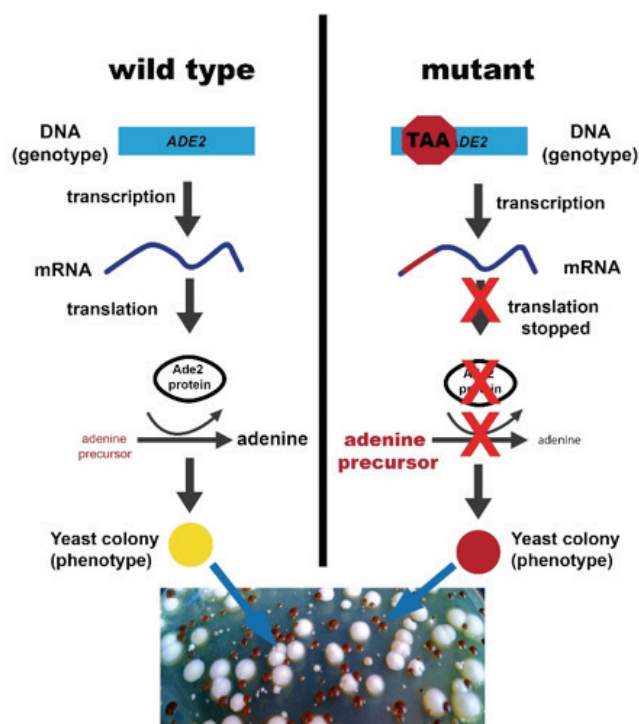


FIGURE 1. Schematic diagram depicting how the introduction of a stop codon by Cas9-mediated genome editing to *ADE2* leads to eventual abrogation of Ade2 translation and buildup of adenine biosynthesis intermediate. Examples of red (edited) and white (not edited) colonies are also shown.

of fuel ethanol and medicines (e.g., insulin). The described protocols use CRISPR to insert a stop codon into the *ADE2* gene (a member of the adenine biosynthesis pathway). Mutations in *ADE2* lead to the buildup of a precursor in the adenine biosynthesis pathway that turns the yeast a bright red color (Fig. 1). (It should be noted that while this exercise targets *ADE2*, any gene could be targeted by CRISPR in *S. cerevisiae* or a number of other yeast species by altering the guide and repair template sequences.) (8)

Transformation of *S. cerevisiae*

Grow overnight yeast extract-peptone-dextrose (YPD) culture of *S. cerevisiae* at room temperature, ideally to Optical Density 600 nm (OD_{600}) less than six. Serial dilutions of overnights are helpful to ensure an overnight is at the appropriate OD when the lab period begins. Higher ODs will still likely transform but will be less efficient. Strains that work well for transformation include, but are not limited to, BY4741 or BY4743 strains classically used in yeast genetic laboratories, brewing strains available at homebrew stores and online (we have found Safbrew T-58, Lallemand Munich wheat, Lallemand Nottingham Ale, and Lallemand BRY-97 work well), and baking strains available at grocery stores and online (we have found Red Star Yeast works well). Multiple transformations with a variety of strains can be performed to compare transformation and gene editing rates between

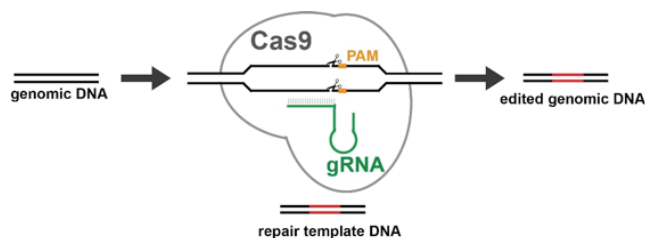


FIGURE 2. Cartoon depiction of the roles of PAM sites, Cas9, repair templates, and guide RNAs in CRISPR-mediated genome editing.

strains. Transform aliquots of yeast with either plasmid pVI382 (8), which encodes Cas9 and RNA guides, or plasmid pVG1 (8), which also contains repair templates designed to introduce stop codons to *ADE2* (Fig. 3) (both plasmids are available through Addgene). Protocols to perform LiAc yeast transformations in under two hours, all reagent recipes, and product numbers are provided in the supplemental materials (Appendices 1, 2, and 3).

Plate each transformation on a separate petri dish containing appropriate media. If using *ura-* strains, such as BY4741, plate onto media lacking uracil Synthetic Complete–Uracil (SC–Ura). If using a prototrophic strain, such as a brewing or baking strain, plate onto YPD containing 100 μ g/mL nourseothricin. *URA3* and nourseothricin resistance (Nat^R) markers are encoded on the plasmid (Fig. 3). Edited colonies will appear and accumulate red color over the next two to seven days, while unedited colonies will remain white (Fig. 1). Students will count colonies and record their results during this time or next class. Suggested pre-lab exercises, learning objectives, post lab exercises, as well as all media recipes are provided in the supplemental materials (Appendices 1, 2, and 3).

Pedagogical aspects of activities

Activities and assessments provided in Appendix 1 will guide students through the process of genetically engineering yeast. Reading the protocol and pre-lab questions will teach students the fundamentals of CRISPR-mediated genome editing in *S. cerevisiae*. Many of these core strategies are applicable to other model systems that are generally too complex to use in an undergraduate laboratory exercise. Furthermore, during the exercise, students will learn a number of microbiology techniques, such as transformation, that can be applied to analogous activities, for example transformation of other organisms. This allows a CRISPR transformation lab to replace or reinforce such labs in a curriculum. Finally, focused relevant discussion of genome editing during the incubation steps of this laboratory provide excellent opportunities for students and instructors to investigate current advances in genome editing. As CRISPR mediated genome editing is a rapidly developing field, such discussions will provide investigators and students an opportunity to explore cutting-edge technologies highly relevant to a wide variety of careers.

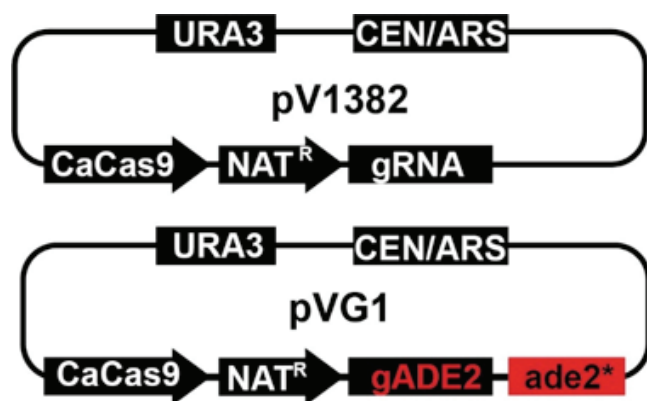


FIGURE 3. Cartoon maps of the plasmids used for these exercises. *CEN/ARS*, yeast centromere sequence enables replication of the plasmid in *S. cerevisiae*; *URA3*, auxotrophic marker, allows BY4741, which normally cannot grow without uracil, to grow without uracil; *CaCas9*, gene encoding the Cas9 nuclease, *NAT^R*, antibiotic resistance marker, allows for yeast to grow in the presence of the antibiotic nourseothricin; *gRNA*, site where a guide RNA could be cloned; *gADE2*, guide RNA targeting *ADE2*; *ade2**, repair template that places a stop codon in *ADE2*.

Student achievement is readily assessable using the materials provided in Appendix I. For instance, pre-lab and post-lab reports and data tables can be collected and used to assess mastery. In addition, student participation during discussions could be used to assess understanding. However, the materials we provide are a guide, and we encourage instructors to edit these materials or create materials they feel are appropriate for their class to assess learning.

Tips for improving transformation

Although transformation with 5 µg of plasmid will, in most cases, give a number of correct transformants, additional plasmid can be used if more colonies are desired. Maxi and mega preps are excellent, cost-effective ways to generate large amounts of plasmid. In addition, allowing cells to recover after heat shock for longer can increase the number of transformants recovered. Finally, electroporation is often more efficient than heat shock transformation but requires special equipment. Protocols for electroporation can be found in Vyas *et al.* (8).

Safety issues

The experiments described in this paper genetically modify *S. cerevisiae*. In addition, plasmid must be purified for the experiments from *E. coli*. In the United States the work described is at Biosafety Level I (BSL1). If working in a different country, the laws guiding work with genetically modified organisms could differ and should be followed. Instructors and students should follow the ASM Biosafety Guidelines (<https://www.asm.org/Guideline/ASM-Guidelines-for-Biosafety-in-Teaching-Laborator>).

CONCLUSIONS

The above-described experimental design and supplemental materials provide students with the opportunity to gain an appreciation for how changes in genotype can lead to changes in phenotype while gaining experience in cutting-edge genome editing technology. The experiments offered are suitable for an introductory undergraduate or advanced placement (AP) high school biology course as they can be performed in two hours, do not require expensive equipment, are easily scalable for large classes, and provide easily interpretable results. These experiments can be embellished by having students design guide sequences or repair templates to edit any *S. cerevisiae* gene of interest.

SUPPLEMENTAL MATERIALS

- Appendix I: Laboratory exercises
- Appendix 2: Media recipes
- Appendix 3: Answer key

ACKNOWLEDGMENTS

This work was supported by R15AI130950 to DAB. The authors declare they have no conflicts of interest.

REFERENCES

1. Lodish HF, Berk A, Kaiser C, Krieger M, Scott MP, Bretscher A, Ploegh HL, Matsudaira PT. 2008. Molecular cell biology, 6th ed. W.H. Freeman, New York.
2. Cong L, Ran FA, Cox D, Lin S, Barretto R, Habib N, Hsu PD, Wu X, Jiang W, Marraffini LA, Zhang F. 2013. Multiplex genome engineering using CRISPR/Cas systems. *Science* 339:819–823.
3. Jinek M, Chylinski K, Fonfara I, Hauer M, Doudna JA, Charpentier E. 2012. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science* 337:816–821.
4. Hsu PD, Lander ES, Zhang F. 2014. Development and applications of CRISPR-Cas9 for genome engineering. *Cell* 157:1262–1278.
5. Bolotin A, Quinquis B, Sorokin A, Ehrlich SD. 2005. Clustered regularly interspaced short palindrome repeats (CRISPRs) have spacers of extrachromosomal origin. *Microbiology* 151:2551–2561.
6. Shah SA, Erdmann S, Mojica FJ, Garrett RA. 2013. Protospacer recognition motifs: mixed identities and functional diversity. *RNA Biol* 10:891–899.
7. DiCarlo JE, Norville JE, Mali P, Rios X, Aach J, Church GM. 2013. Genome engineering in *Saccharomyces cerevisiae* using CRISPR-Cas systems. *Nucleic Acids Res* 41:4336–4343.
8. Vyas VK, Bushkin GG, Bernstein DA, Getz MA, Sewastianik M, Barrasa MI, Bartel DP, Fink GR. 2018. New CRISPR mutagenesis strategies reveal variation in repair mechanisms among fungi. *mSphere* 3: doi 10.1128/mSphere.00154-18.